

Practical #6: Practice question on Maven + Docker Integration

Statement: “Containerizing a Java Web Application for Cloud Deployment” You are working as a DevOps engineer in a startup. The development team has built a Java web application (WAR file) using Maven. Your task is to:

- Build the WAR using Maven
- Containerize the application using Docker
- Deploy it on a containerized environment (simulating cloud deployment)

```

ICTUS@Jarvis MINGW64 ~/web-demo0 (master)
$ ls
pom.xml  src/  target/

ICTUS@Jarvis MINGW64 ~/web-demo0 (master)
$ cd src

ICTUS@Jarvis MINGW64 ~/web-demo0/src (master)
$ ls
main/

ICTUS@Jarvis MINGW64 ~/web-demo0/src (master)
$ cd main

ICTUS@Jarvis MINGW64 ~/web-demo0/src/main (master)
$ ls
resources/  webapp/

ICTUS@Jarvis MINGW64 ~/web-demo0/src/main (master)
$ cd webapp

ICTUS@Jarvis MINGW64 ~/web-demo0/src/main/webapp (master)
$ ls
WEB-INF/  index.jsp

ICTUS@Jarvis MINGW64 ~/web-demo0/src/main/webapp (master)

```

STEP-01: Create the template & get inside the folder

[illegible]

STEP-02: Edit the pom.xml and make a docker file

STEP-03: mvn clean install

```
MINGW64/C:/Users/VICTUS/web-demo
Downloaded from central: https://repo.maven.apache.org/maven2/commons-cli/commons-cli/1.2/commons-cli-1.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-war-runner/2.2/tomcat7-war-runner-2.2.jar (25 kb at 15 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/2.4.2/maven-archiver-2.4.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/httpcomponents/httpcore/4.3/httpcore-4.3.jar (22 kb at 175 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.13/plexus-interpolation-1.13.jar
Downloaded from central: https://repo.maven.apache.org/maven2/commons-cli/commons-cli/1.2/commons-cli-1.2.jar (41 kb at 21 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-classworlds/2.2.2/plexus-classworlds-2.2.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/2.4.2/maven-archiver-2.4.2.jar (20 kb at 10 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.2/commons-io-2.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.2/commons-io-2.2.jar (61 kb at 31 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.4.1/commons-compress-1.4.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/httpcomponents/httpclient/4.3.1/httpclient-4.3.1.jar (586 kb at 276 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.2/commons-io-2.2.jar (174 kb at 77 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/2.1.1/plexus-archiver-2.1.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-classworlds/2.2.2/plexus-classworlds-2.2.2.jar (46 kb at 20 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/junit/junit/4.10/junit-4.10.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/tukaani/xz/1.0/xz-1.0.jar (95 kb at 41 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/hamcrest/hamcrest-core/1.1/hamcrest-core-1.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/com/google/guava/guava/10.0.1/guava-10.0.1.jar (1.5 MB at 651 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/classworlds/classworlds/1.1-alpha-2/classworlds-1.1-alpha-2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/2.1.1/plexus-archiver-2.1.1.jar (185 kb at 76 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/2.0.3/plexus-io-2.0.3.jar
Downloaded from central: https://repo.maven.apache.org/maven2/classworlds/classworlds/1.1-alpha-2/classworlds-1.1-alpha-2.jar (38 kb at 15 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0.15/plexus-utils-3.0.15.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/hamcrest/hamcrest-core/1.1/hamcrest-core-1.1.jar (77 kb at 31 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-filtering/1.0/maven-filtering-1.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/reporting/maven-reporting-api/2.2.1/maven-reporting-api-2.2.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.4.1/commons-compress-1.4.1.jar (241 kb at 97 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/junit/junit/4.10/junit-4.10.jar (253 kb at 101 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/doxia/doxia-logging-api/1.1/doxia-logging-api-1.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/2.0.3/plexus-io-2.0.3.jar (38 kb at 24 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interactivity-api/1.0-alpha-4/plexus-interactivity-api-1.0-alpha-4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/reporting/maven-reporting-api/2.2.1/maven-reporting-api-2.2.1.jar (9.8 kb at 3.9 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/doxia/doxia-sink-api/1.1/doxia-sink-api-1.1.jar (13 kb at 5.1 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/doxia/doxia-logging-api/1.1/doxia-logging-api-1.1.jar (11 kb at 4.4 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-build-api/0.0.4/plexus-build-api-0.0.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0.15/plexus-utils-3.0.15.jar (239 kb at 93 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/slf4j/slf4j-api/1.7.5/slf4j-api-1.7.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/slf4j/jcl-over-slf4j/1.7.5/jcl-over-slf4j-1.7.5.jar (13 kb at 5.2 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus-build-api/0.0.4/plexus-build-api-0.0.4.jar (5.8 kb at 2.6 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/slf4j/jcl-over-slf4j/1.7.5/jcl-over-slf4j-1.7.5.jar (17 kb at 6.4 kb/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/slf4j/slf4j-api/1.7.5/slf4j-api-1.7.5.jar (26 kb at 10 kb/s)
[INFO] Running war on http://localhost:8080/
[INFO] Creating Tomcat server configuration at C:/Users/VICTUS/web-demo/target/tomcat
[INFO] create webapp with contextPath
Mar 31, 2026 12:46:14 PM org.apache.coyote.AbstractProtocol init
INFO: Initializing ProtocolHandler ["http-bio-8080"]
Mar 31, 2026 12:46:14 PM org.apache.catalina.core.StandardService startInternal
INFO: Starting service Tomcat
Mar 31, 2026 12:46:14 PM org.apache.catalina.core.StandardEngine startInternal
INFO: Starting Servlet Engine: Apache Tomcat/7.0.47
Mar 31, 2026 12:46:15 PM org.apache.coyote.AbstractProtocol start
INFO: Starting ProtocolHandler ["http-bio-8080"]
```

STEP-04: Run the container



Hello! This webpage is running using Maven.

STEP-05: Open your browser and visit:

<http://localhost:8080/>